

Math 465: Homework 3

Due: Tuesday, September 22, by 11:59PM

Instructions. Write full and complete solutions to the following problems and submit them on Gradescope by the deadline. For each problem, explain the reasoning behind your answer; a formula or numerical answer without justification is not a complete solution. **If your final answer is numerical, please carry out the arithmetic** and state the resulting number (so, for example, if your answer is $6! - 5!$, I want you to also tell me the answer is 600.)

You may discuss the problems with other students, but you must write up your solutions independently and in your own words. You are not allowed to use generative AI tools, Internet searches, or any other external means to directly obtain a solution. Generative AI tools may also not be used to proofread your writing or L^AT_EX. If you are unsure about what is or is not permitted, please ask.

At the beginning of your submission, acknowledge any permitted books, notes, or other sources you consulted, and list the names of any students with whom you discussed the homework.

1. Find a closed-form expression for the generating function of each of the following sequences, indexed by $n \geq 0$.

(a) $a_n = 4^n + 2 \cdot 3^n$.

(b) $b_n = \binom{n+2}{2} 2^n$.

2. For $n \geq 0$, let a_n be the number of bags containing n pieces of fruit, chosen from apples, oranges, bananas, and pears, subject to the following restrictions: the number of apples is even, there are at most two oranges, the number of bananas is a multiple of three, and there is at most one pear. Fruits of the same type are indistinguishable, and their order in the bag does not matter.

- (a) Find the generating function

$$\sum_{n \geq 0} a_n x^n.$$

- (b) Simplify your generating function and use it to obtain an explicit formula for a_n .

3. Fix integers $k \geq 1$ and $0 \leq r \leq k$. For $n \geq 0$, let a_n be the number of weak compositions of n into k parts in which exactly r parts are odd.

- (a) Find the generating function

$$\sum_{n \geq 0} a_n x^n.$$

- (b) Obtain an explicit formula for a_n . When is it zero?

4. Let $r, s \geq 1$ and $n \geq 0$ be integers. Prove that

$$\sum_{j=0}^n \binom{r+j-1}{r-1} \binom{s+n-j-1}{s-1} = \binom{r+s+n-1}{r+s-1}.$$

Give two proofs: one using generating functions and one by counting weak compositions.

5. Fix integers $k \geq 1$ and $b \geq 0$. For $n \geq 0$, let a_n be the number of weak compositions

$$u_1 + \cdots + u_k = n$$

satisfying $u_i \leq b$ for every i .

- (a) Find the generating function

$$\sum_{n \geq 0} a_n x^n.$$

- (b) Use the binomial theorem to obtain a formula for a_n in terms of ordinary binomial coefficients. Specify the range of summation. Your final answer should not be left as “the coefficient of x^n ” in a generating function.

Hint: Write each finite geometric series as a quotient, and expand the numerator using the binomial theorem.

6. Recall that the code of a permutation $w \in S_n$ is the sequence $(c_1, \dots, c_n)$, where c_k is the number of entries smaller than k that occur to the right of k .

Fix integers $n \geq 1$ and $0 \leq r \leq n - 1$. Consider the permutations in S_n in which every entry has at most r smaller entries to its right.

- (a) Find the generating polynomial for these permutations with respect to their number of inversions. That is, let a_j be the number of permutations in S_n having exactly j inversions and satisfying the restriction above. Find a product formula for the generating polynomial

$$\sum_{j \geq 0} a_j x^j.$$

- (b) Find the total number of these permutations.

- (c) In the case $n \geq 2$ and $r = 1$, find an explicit formula for the number of these permutations having exactly j inversions, for each integer $j \geq 0$.

7. Write $[n] = \{1, \dots, n\}$. For a permutation $w \in S_n$, let w^{-1} denote its inverse as a function. The *Rothe diagram* of w is the following set of cells in an $n \times n$ grid:

$$D(w) = \{(i, j) \in [n] \times [n] : j < w(i) \text{ and } i < w^{-1}(j)\}.$$

Rows are numbered from top to bottom, and columns from left to right. The *transpose* of a diagram D is

$$D^T = \{(j, i) : (i, j) \in D\}.$$

- (a) Draw $D(w)$ for $w = 35142$.

- (b) Prove that $D(w^{-1}) = D(w)^T$.

- (c) Give a bijection between the cells of $D(w)$ and the inversions of w . Deduce that

$$\text{inv}(w) = \text{inv}(w^{-1}).$$

8. A *descent* of a permutation $w = w_1 \cdots w_n \in S_n$ is a position $i \in [n - 1]$ such that $w_i > w_{i+1}$. Define the *descent set* and the *major index* by

$$\text{Des}(w) = \{i \in [n - 1] : w_i > w_{i+1}\}, \quad \text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

- (a) Compute $\text{Des}(w)$, $\text{maj}(w)$, and $\text{inv}(w)$ for every $w \in S_3$. Give an example for which $\text{maj}(w) \neq \text{inv}(w)$.
- (b) Let $n \geq 2$ and fix $w \in S_{n-1}$. Insert n into each of the n possible positions. Prove that the resulting increases in major index are

$$0, 1, \dots, n-1,$$

each occurring once.

- (c) Deduce that, for every $n \geq 1$,

$$\sum_{w \in S_n} x^{\text{maj}(w)} = \prod_{k=1}^n (1 + x + \dots + x^{k-1}).$$

9. Use the cycle-insertion encoding $(b_1, \dots, b_n)$ from class. We build the permutation by inserting $1, 2, \dots, n$ in order: $b_k = 0$ means that k starts a new cycle, while $b_k = j > 0$ means that k is inserted immediately after j in its current cycle.

- (a) Find the encoding of

$$w = (1, 4, 6)(2, 5)(3),$$

and reconstruct the permutation encoded by

$$(0, 1, 0, 2, 0, 3).$$

- (b) Prove that $b_k = 0$ exactly when k is the smallest element of its cycle in the final permutation.
- (c) Given $S \subseteq [n]$, find the number of permutations for which the set of smallest elements of the cycles is exactly S . Include the case $1 \notin S$.

10. A permutation $w \in S_n$ is an *involution* if $w(w(i)) = i$ for every $i \in [n]$. A *nontrivial involution* is an involution other than the identity permutation.

- (a) Prove that a permutation is an involution exactly when all its cycles have length 1 or 2.
- (b) Let I_n be the number of involutions in S_n , including the identity, and set $I_0 = 1$. Prove that

$$I_n = I_{n-1} + (n-1)I_{n-2} \quad (n \geq 2),$$

and determine I_1 .

- (c) Prove that the number of nontrivial involutions in S_n is odd whenever $n \geq 2$. Give a proof by pairing permutations with their inverses.